

PERN

Pollution & Environmental Risk Navigator

Full-stack IoT Environmental Intelligence Platform

Real-time monitoring · AI analysis · Multi-protocol IoT · Enterprise security

PostgreSQL 15

Express 5

React 19

Node.js

TypeScript 6

MQT

WebSocket

Logto OIDC

OpenRouter AI

Docker

50

Page Views

14

Sensor Types

10

Virtual Sensors

260

Automated Tests

Version 3.1 · Project Documentation · August 2026

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STEM Gharbiya · Grade 11 · 2026

Contents

• Executive Summary	3
• Project Overview	4
• System Architecture	5
• Technology Stack	6
• Feature Set	8
• Sensor Suite	10
• Database Schema	11
• API Endpoints	12
• Backend Module Map	13
• Frontend Module Map	14
• Security Model	15
• AI Intelligence & Notifications	16
• Configuration	17
• Testing & Quality	18
• Copyright, License & Ownership	19

PERN (Pollution & Environmental Risk Navigator) is a full-stack IoT environmental intelligence platform for the Nile Delta region. It ingests live sensor data over MQTT, computes scientific environmental indices, and surfaces real-time intelligence, AI analysis, alerts, automation and reporting through a modern web dashboard.

At a glance

PERN is built on the PERN stack — PostgreSQL 15, Express 5, React 19 and Node.js — containerized as a 5-service Docker Compose stack, secured with Logto OIDC authentication, RBAC, rate limiting and Helmet hardening, and powered by an OpenRouter LLM with a 1M-token context.

50**Page Views**

Complete monitoring surface

14**Sensor Types**

Air, water, soil, thermal

10**Virtual Sensors**

AQI, WQI, EHI, risk + more

5**Protocols**

MQTT, REST, WS, CoAP, LoRaWAN

23**DB Tables**

PostgreSQL data model

260**Tests**

222 backend + 38 frontend

Highlights

- Live dashboard with EHI scoring, alert counts and device status across 50 page views.
- AI copilot: chatbot (SSE streaming), time-series predictions with confidence intervals, root-cause analysis, plain-English rule generation, and a knowledge base.
- Soft-sensor estimation with leave-one-out validation (R^2 , MAE, RMSE, MBE) across four unlock tiers.
- Server-side automation engine with a 30-second cooldown and real ntfy push notifications.
- Global data integration: OpenAQ, WAQI, Sensor.Community, NASA FIRMS and Sentinel-5P CAMS (Open-Meteo) with live compliance tracking by country and framework.
- Notification Center with channel health, per-user preferences and a dispatch-history console.
- Fully internationalized (English / Arabic) with a dark glassmorphism design system.

2 Project Overview

PERN delivers environmental health intelligence in the Nile Delta region by combining distributed IoT sensing with scientific modelling and AI. Physical devices (ESP32, NodeMCU, gateways) stream measurements; the platform validates, stores and enriches them into actionable indices, alerts and forecasts.

Why it matters

- Real-time visibility of air, water and soil quality for communities, farms and researchers.
- Virtual sensors fill data gaps where physical instruments are missing or too costly.
- AI reduces time-to-insight — asking questions in natural language instead of querying databases.
- Automation and alerts turn raw measurements into immediate, channel-agnostic notifications.

Design principles

Principle	How PERN applies it
Separation of concerns	Frontend / backend / broker / auth / data are independent services
Security first	AuthN + AuthZ, parameterized SQL, sanitization, rate limits, helmet
Scientific grounding	Algorithms follow WHO, EPA, NSF, FAO-56, ISO, CIE standards
Realtime by default	MQTT ingestion and WebSocket delivery out of the box
Testability	260 Vitest tests plus strict TypeScript type checking
Tenant isolation	Multi-organization data separation with role-scoped access

3 System Architecture

Logical architecture

IoT & Devices



Protocol Gateway



Backend Platform



Data & Services



Devices publish over MQTT; the backend persists readings, evaluates thresholds and automation rules, and pushes realtime updates to the React dashboard over REST and WebSocket.

Data pipeline



Deployment (Docker Compose)

Service	Container	Exposed Ports
Frontend	React + Vite (Nginx)	80
Backend	Express + Node.js	3000
Database	PostgreSQL 15	5432
MQTT Broker	Mosquitto	1883 / 9001
Auth	Logto	3001 / 3002

```
docker compose up -d --build
```

4 Technology Stack

Frontend dependencies

Package	Version	Purpose
react / react-dom	^19.2.7	UI runtime, concurrent rendering
react-router-dom	^7.18.1	Routing with lazy-loaded pages
typescript	~6.0.2	Strict typing across 50 pages
vite	^8.1.1	Dev server and production build
tailwindcss	^3.4.17	Dark glassmorphism design system
recharts	^3.9.2	Composable SVG charts
leaflet / react-leaflet	^1.9.4 / ^5.0.0	Global sensor maps + clusters
framer-motion	^12.42.2	Page / section animations
react-markdown + remark-gfm	^10.1.0 / ^4.0.1	Knowledge base rendering
jspdf	^4.2.1	Client-side PDF report generation
xlsx	^0.18.5	Excel-compatible export
mqtt	^5.15.2	WebSocket MQTT client
@logto/browser	^3.0.13	Logto OIDC authentication
lucide-react	^1.25.0	Icon set
mermaid	^11.16.0	Diagram rendering

Backend dependencies

Package	Version	Purpose
express	^5.2.1	HTTP API framework
pg	^8.22.0	PostgreSQL client
mqtt	^5.15.2	MQTT broker connection
ws	^8.18.3	WebSocket server (8081)
helmet	^8.3.0	Security headers (CSP, HSTS)
jose	^6.2.3	Logto JWT verification
nodemailer	^9.0.3	SMTP email notifications
pdfkit	^0.19.1	Server-side PDF generation
chartjs-node-canvas	^5.0.0	Server-side chart rendering
dotenv	^17.4.2	Environment configuration
cors	^2.8.6	Origin allowlist policy
compression	^1.8.1	Response compression
node-fetch	^2.7.0	HTTP client for external APIs

Tooling & DevOps

- Docker Compose (5 services) · Vitest (test runner) · oxlint (linter) · tsc --noEmit (type check).

Recent enhancement areas

References — scientific standards library with BibTeX/CSV export; Research — OLS regression, t-tests and lagged cross-correlation; Virtual Sensors — leave-one-out validation and scenario simulation; Alerts — analytics, history and full rule management.

Real-Time Monitoring

- Live dashboard: sensor readings, EHI scores, alert counts, device status.
- MQTT streaming from ESP32, NodeMCU and other devices.
- Global sensor maps with marker clustering and plume overlays.
- Device health dashboards, firmware management and over-the-air updates.

Virtual Sensors & Research

- 10 computed indices: AQI, WQI, EHI, Risk Score, Indoor Air Quality, Corrosion Index, BOD Estimate, Thermal Comfort, Agricultural Suitability, Human Exposure.
- Estimation of missing measurements with leave-one-out validation (R^2 , MAE, RMSE, MBE).
- Statistical research tools: OLS regression, two-tailed t-tests, lagged cross-correlation.
- Scenario simulation with heatwave / cold night / humid day / high pollution / water-quality presets.

AI-Powered Intelligence

- Chatbot — natural-language Q&A with SSE streaming answers.
- Predictions — time-series forecasting with confidence intervals.
- Rule Generator — describe automation rules in plain English.
- Root Cause Analysis — AI-driven anomaly investigation.
- Knowledge Base — searchable environmental monitoring documentation.

Alerts & Automation

- Threshold-based alert rules with $>$, $>=$, $<$, $<=$, $==$, $!=$, between, outside operators.
- Severity levels, per-sensor thresholds and acknowledgement workflow.
- Automation engine with a 30-second cooldown and PostgreSQL-persisted rules.
- Real ntfy push notifications on rule triggers.

Multi-Protocol Device Support

Protocol	Status	Use Case
MQTT	Active	Primary IoT protocol (Mosquitto)
HTTP REST	Active	Simple device integrations
WebSocket	Active	Browser-based devices
CoAP	Simulated	Constrained / low-power devices
LoRaWAN	Simulated	Long-range IoT gateways

Export, Reporting & Organizations

- CSV export for readings and alerts; Excel spreadsheets via xlsx.
- PDF report generator (daily summary, water/air quality, risk assessment, compliance, wind, comprehensive).
- Multi-organization support, team invitations, role management and organization-scoped settings.

Physical sensor types

Key	Name	Unit	Safe Range
ph	pH	—	6.5 – 8.5
tds	TDS	ppm	0 – 500
wT	Water Temperature	°C	10 – 30
dO	Dissolved Oxygen	mg/L	5 – 14
tb	Turbidity	NTU	0 – 5
pm25	PM2.5	µg/m³	0 – 35
mq	MQ135 Gas	ppm	0 – 1.0
tmp	Air Temperature	°C	15 – 35
hum	Humidity	%	30 – 70
co2	CO ₂	ppm	300 – 1000
nh3	NH ₃	ppm	0 – 25
voc	VOC	ppb	0 – 500
sm	Soil Moisture	%	20 – 60
light	Light Intensity	lux	0 – 100000

Virtual / estimated sensors

The platform computes environmental indices and estimates missing measurements with leave-one-out validation (R^2 , MAE, RMSE, MBE) across four unlock tiers:

- Tier 1 (temp + humidity) — dew point, heat index, VPD, WBGT, absolute humidity, apparent temperature, frost point, cloud base.
- Tier 2 (+ MQ-135 gas) — NO_x, pollution density, CO₂, proxy, AQI, PM2.5.
- Tier 3 (+ light) — UV index, solar radiation, evapotranspiration, PPFD.
- Tier 4 (water sensors) — dissolved oxygen, pH, NH₃, turbidity.

Scientific grounding

All thresholds and equations reference real standards: WHO Air Quality Guidelines (2021), EPA NAAQS/AQI, NSF WQI, FAO-56 Penman-Monteith, ASHRAE 62.1, ISO 7243, CIE photometry (085/087), Magnus/Alduchov psychrometrics, and peer-reviewed statistical time-series methods.

PostgreSQL 15 with 30 tables, parameterized queries throughout, and organization-scoped data isolation. Tables are grouped by domain:

Ingestion

- sensor_readings · device_readings · devices · device_api_keys · device_metadata · device_health

Alerts & Automation

- alerts · alert_rules · alert_history · alert_thresholds · automation_rules · automation_logs · notification_preferences

Environment

- ehi_history · actuator_commands · data_retention_policies

Organizations & Users

- organizations · team_members · users · audit_logs

AI & Firmware

- ai_conversations · ai_messages · firmware_versions

Global Data Fabric

- global_data_sources · virtual_sensors · external_readings · sensor_confidence_scores · compliance_frameworks · wind_trajectories · plume_events · global_api_keys

Data integrity measures

- SENSOR range validation and unit normalization on ingest.
- Data retention policies with server-side cleanup.
- Audit logging for administrative and device actions.
- EHI history snapshots for trend analysis.

8 API Endpoints

Core data & ingestion

Method	Endpoint	Description
GET	/api/sensors	Recent sensor readings
POST	/api/sensors	Ingest sensor data
POST	/api/readings	Unified reading ingestion
GET	/api/thresholds	Sensor thresholds
POST	/api/thresholds	Save thresholds
POST	/api/export/csv	CSV export (readings / alerts)

Alerts

Method	Endpoint	Description
GET/POST	/api/alerts/rules	List / create alert rules
DELETE	/api/alerts/rules/:id	Delete rule (admin/manager)
GET	/api/alerts/history	Alert history (limit/severity/device)
POST	/api/alerts/history/:id/acknowledge	Acknowledge an alert
GET	/api/alerts/history/stats	Alert analytics

AI, automation & devices

Method	Endpoint	Description
POST	/api/chatbot/chat	AI chat (SSE streaming)
POST	/api/ai-tools/*	AI analysis endpoints
POST	/api/persistence/rules	Persist automation rules
GET/POST	/api/devices	Device management
GET	/api/protocols/status	Protocol adapter status
POST	/api/reports/generate	Generate PDF report
WS	ws://localhost:8081	Realtime actuator updates (JWT)

Organizations & notifications

- Organization CRUD, team invites, member roles, org settings.
- Notification channel configuration (ntfy, email, SMS, Slack webhooks).
- GET /api/notifications/status · GET /api/notifications/log · preferences CRUD + test-send.

Global data & public API

- /api/v3/* global ingestion, /api/export (CSV), /public/v1/* keyed public gateway, /api/keys management.

Route modules (17)

ai-tools · alerts · api-keys · automation-control · chatbot · devices · export · firmware · global-v3 · notifications · organizations · persistence · protocols · public · reports · seed · users

Service modules (26)

ai-analysis · ai-briefing · ai-cache · ai-copilot · ai-references · ai-router · ai-service · alert-engine · analysis-engine · anomaly-detector · automation-engine · compliance-engine · conversation-memory · data-retention · data-normalizer · device-config · global-ingestion · llm-api · llm-data · notification-dispatcher · public-api · root-cause · rule-generator · satellite-engine · trust-engine · wind-engine

Middleware (12)

audit-logger · brute-force · csrf · device-auth · error-handler · query-validator · rate-limiter · rbac · request-logger · sanitize · security-headers · validator

Protocol adapters (7) + WebSocket

mqtt-adapter · http-adapter · websocket-adapter · coap-adapter · lorawan-adapter · protocol-adapter · protocol-manager | websocket/actuator-ws.js

Core files

File	Purpose
<code>server.js</code>	Express app, MQTT gateway, route mounting, automation loop
<code>db.js</code>	PostgreSQL schema, queries, migrations (30 tables)
<code>auth.js</code>	Logto OIDC JWT verification
<code>utils/</code>	Logger and environment validation

10

Frontend Module Map

Pages by domain (50)

Domain	Pages
Monitoring	Dashboard · Sensors · History · Weather · Maps · PlumeMap · RealSensorMap
AI & Analysis	AI · Chatbot · Predictions · Research · RuleGenerator · Resources
Virtual Sensors	VirtualSensors · CompareVirtualSensors · RealDataValidation
Alerts & Automation	Alerts · Automation · Compliance
Devices & IoT	DeviceHub · DeviceDetail · Devices · DeviceConnection · DeviceLifecycle · DeviceHealthDashboard · DeviceSetupGuide · Firmware
Global Data	GlobalSensorsV3 · DataSources · ProtocolStatus
Notifications	NotificationsCenter
Reports & Docs	Reports · References · Knowledge · KnowledgeHub
Organization	Analytics · SecurityAudit · TrustDashboard · DigitalTwin · SystemStatus
Platform	ApiConsole · ConnectionTest · SensorCalibration · Settings · Support · Showcase
Auth	Login · AuthCallback

Core libraries (lib/)

```
api-client · mqtt-client · ws-client · data-provider · auth-context · i18n · virtual-sensors · virtual-sensor-estimators · virtual-sensor-unified · scientific-core · scientific-ehi · ai-predict-engine · recommendation-engine · trust/confidence-scoring · openaq-service · sensor-estimator · device-profiles · esp32-sketch-generator · audit-log · types · roles
```

Reusable components (19)

- ui.tsx (15+ memoized primitives) · charts.tsx · ResponsiveTable · DeviceSelector · DateRangePicker
- GlobalSensorMap · SignalFlow · AnimatedBackground · CommandPalette · Toast · Skeleton · LoadingSpinner
- RequireAuth · PageErrorBoundary · ErrorBoundary · MobileBottomNav · LanguageSwitcher · OrganizationSwitcher · OnboardingModal

11 Security Model

Authentication & authorization

- Logto OIDC — standards-compliant JWT verification via jose.
- RBAC — four roles: admin, manager, member, viewer.
- Ownership checks — users can only modify their own resources.
- WebSocket JWT verification on upgrade handshake; tokens stored in sessionStorage.

API hardening

- Sliding-window rate limiting per IP with configurable per-route limits.
- Rate-limit headers: X-RateLimit-Limit, X-RateLimit-Remaining, Retry-After.
- Input validation (sensor ranges, automation rules) and XSS-pattern sanitization on all bodies and query params.
- Request logging — method, URL, status, duration, IP, user-agent.

Security headers (Helmet)

- Content Security Policy with restrictive directives.
- HSTS (2 years, includeSubDomains, preload).
- X-Frame-Options: DENY · X-Content-Type-Options: nosniff · cross-origin policies.

Environment & CORS

- Explicit origin allowlist — no wildcard origins; restricted allowedHeaders.
- Parameterized SQL everywhere — no injection surface.
- NODE_ENV-controlled error exposure; stack traces hidden in production.
- Dedicated middleware: audit-logger, brute-force, csrf, device-auth, sanitize, security-headers.

12 AI Intelligence & Notifications

AI stack

Component	Implementation
Provider	OpenRouter API
Model	nvidia/nemotron-3-super-120b-a12b:free
Context window	1M tokens
Streaming	SSE (Server-Sent Events) to the chatbot UI
Services	ai-service · llm-client · ai-router · ai-copilot · ai-analysis · root-cause · rule-generator
Memory	Conversation memory with cache (ai-cache)
Validation	AI answer validation + confidence scoring

Capabilities

- Chatbot: natural-language Q&A over live sensor data with streamed answers.
- Predictions: time-series forecasting with confidence intervals.
- Root Cause Analysis: AI-driven anomaly investigation.
- Rule Generator: plain-English automation rule creation.
- Satellite & wind engines, trust scoring and compliance briefing as supporting AI services.

Notification channels

Channel	Use Case
ntfy	Push notifications to devices/browsers (active)
SMTP (nodemailer)	Email alerts to operators
SMS	Emergency SMS delivery
Slack webhooks	Team workspace integration

Alerts are dispatched by the notification-dispatcher service with severity-aware routing. A Notification Center dashboard exposes channel health, per-user preference toggles and a live dispatch-history console (GET `/api/notifications/status` and `/api/notifications/log`).

13 Configuration

Backend environment

Variable	Required	Description
DATABASE_URL	Yes	PostgreSQL connection string
MQTT_BROKER	Yes	MQTT broker URL
OPENROUTER_API_KEY	AI	OpenRouter API key
ENFORCE_AUTH	Prod	true != require Logto login
LOGTO_ENDPOINT	Auth	Logto tenant URL
LOGTO_APP_ID	Auth	Logto application ID
NODE_ENV	Prod	production != hardened mode
ALLOWED_ORIGINS	Prod	Comma-separated CORS origins

Frontend environment

Variable	Description
VITE_LOGTO_ENDPOINT	Logto tenant URL
VITE_LOGTO_APP_ID	Logto application ID
VITE_API_URL	Backend API URL (default: /api)

Local development

```
cd pern-frontend && npm install && npm run dev # Vite dev server
cd pern-backend && npm install && npm start # Express API on :3000
docker compose up -d --build # full stack (5 services)
```

Supported roles & permissions

Role	Capabilities
admin	Full platform access, org management, all settings
manager	Alert rules, thresholds, devices, reports
member	Data viewing, alerts, exports
viewer	Read-only access

14 Testing & Quality

Both codebases ship Vitest suites covering the AI pipeline, virtual-sensor algorithms, data integrity, recommendation engine, alert rules and the full API surface. The frontend is additionally gated by strict TypeScript and lint checks.

```
cd pern-backend && npm test          # 222 tests · 10 files
cd pern-frontend && npm test          # 38 tests · 4 files
cd pern-frontend && npm run typecheck # tsc --noEmit
cd pern-frontend && npx oxlint src    # 0 errors
cd pern-frontend && npm run build     # type-check + production build
```

What is covered

- Virtual sensor estimators — leave-one-out validation, MAE/RMSE/MBE/R².
- AI pipeline — services, validation, confidence scoring, recommendation engine.
- Alert rules — operator evaluation and threshold logic.
- API surface — route handlers, auth, rate limiting, data integrity.
- Global data fabric — source adapters, trust engine, wind/plume, public API gateway, satellite, retention.

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All scientific thresholds and equations reference real standards: WHO AQG 2021, EPA NAAQS/AQI, NSF WQI, FAO-56, ASHRAE 62.1, ISO 7243, CIE 085/087, and peer-reviewed psychrometric and time-series methods.